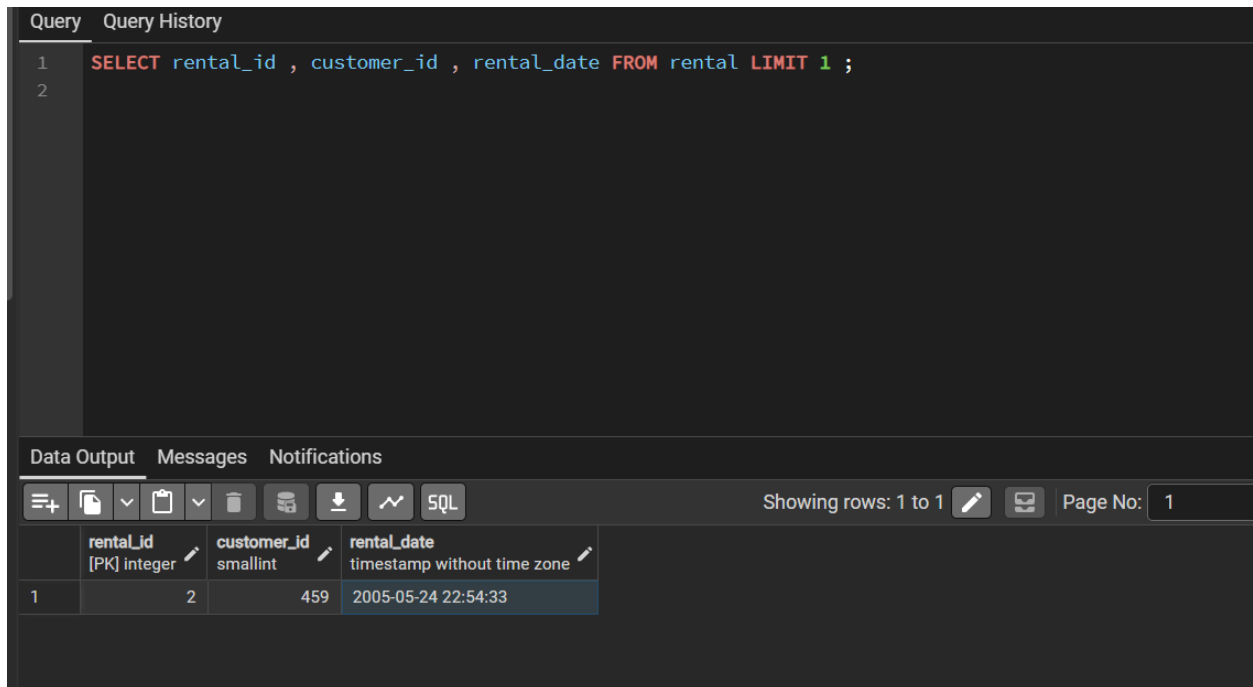


Lab 1 : Transactions & ACID Properties

Tìm rental_id thật để test



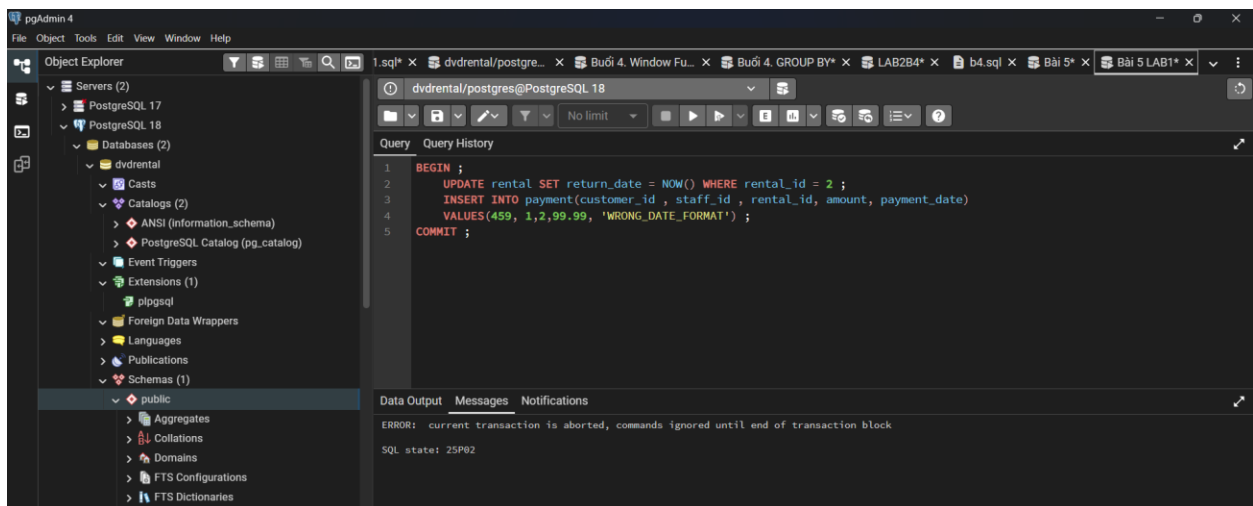
Query Query History

```
1 SELECT rental_id , customer_id , rental_date FROM rental LIMIT 1 ;
2
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

	rental_id [PK] integer	customer_id smallint	rental_date timestamp without time zone
1	2	459	2005-05-24 22:54:33



pgAdmin 4

Object Explorer

- Servers (2)
 - PostgreSQL 17
 - Databases (2)
 - dvdrental
 - Cast
 - Catalogs (2)
 - ANSI (information_schema)
 - PostgreSQL Catalog (pg_catalog)
 - Event Triggers
 - Extensions (1)
 - plpgsql
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries

Query Query History

```
1 BEGIN ;
2 UPDATE rental SET return_date = NOW() WHERE rental_id = 2 ;
3 INSERT INTO payment(customer_id , staff_id , rental_id, amount, payment_date)
4 VALUES(459, 1,2,99.99, 'WRONG_DATE_FORMAT') ;
5 COMMIT ;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Câu INSERT báo lỗi , transaction bị fail , toàn bộ thay đổi bị hủy

The screenshot shows the pgAdmin 4 interface. The top toolbar includes icons for file operations, query execution, and settings. The main query editor displays the following SQL code:

```
1 SELECT rental_id, return_date
2 FROM rental
3 WHERE rental_id = 2;
```

Below the query editor, the 'Data Output' tab is active, showing the results of the query in a table format:

rental_id	return_date
1	2005-05-28 19:40:33

The status bar at the bottom indicates 'Showing rows: 1 to 1' and 'Page No: 1 of 1'.

Return vẫn null -> Atomicity hoạt động đúng.

The screenshot shows the pgAdmin 4 interface with the 'Object Explorer' on the left. The main query editor displays the following SQL code:

```
1
2 ROLLBACK;
3
4 BEGIN;
5
6 UPDATE rental
7 SET return_date = NOW()
8 WHERE rental_id = 2;
9
10 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
11 VALUES (459, 1, 2, 25.00, NOW());
12
13 COMMIT;
```

Below the query editor, the 'Data Output' tab is active, showing the results of the transaction:

WARNING: there is no transaction in progress
COMMIT
Query returned successfully in 153 msec.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Triggers
- category
- city
- Columns
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- country
- customer
- film
- film_actor
- film_category
- inventory
- language
- payment
- rental
- rental_log
- staff
- store
- Trigger Functions (4)
 - check_rental_limit()
 - last_updated()
 - log_rental_action()

Query

```
--SELECT rental_id , customer_id , rental_date FROM rental LIMIT 1 ;
--SELECT rental_id, return_date
--FROM rental
--WHERE rental_id = 2;
SELECT rental_id, return_date
FROM rental
WHERE rental_id = 2;
SELECT +
FROM payment
WHERE rental_id = 2
ORDER BY payment_date DESC
LIMIT 5;
```

Data Output

payment_id [PK] integer	customer_id smallint	staff_id smallint	rental_id integer	amount numeric (5,2)	payment_date timestamp without time zone
32099	459	1	2	25.00	2026-09-18 15:12:29.292667

Showing rows: 1 to 1 Page No: 1 of 1

Return_date kiểm tra xem còn giá trị null không

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Triggers
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- Indexes
- RLS Policies
- Rules
- Triggers
- country
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- film
- film_actor
- film_category
- inventory
- language
- payment
- rental
- rental_log
- staff
- store
- Trigger Functions (4)
 - check_rental_limit()
 - last_updated()
 - log_rental_action()
 - set_rental_date()

Query

```
--SELECT rental_id, return_date
--FROM rental
--WHERE rental_id = 2;
SELECT rental_id, return_date
FROM rental
WHERE rental_id = 2;
SELECT +
FROM payment
WHERE rental_id = 2
ORDER BY payment_date DESC
LIMIT 5;
SELECT rental_id, return_date
FROM rental
WHERE rental_id = 2;
```

Data Output

rental_id [PK] integer	return_date timestamp without time zone
2	2026-09-18 15:12:29.292667

Showing rows: 1 to 1 Page No: 1 of 1

